DOES THE RETIREMENT CONSUMPTION PUZZLE DIFFER ACROSS THE DISTRIBUTION?

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Abstract

Previous research has repeatedly found a puzzling one-time drop in consumption at retirement at the mean or median. This study expands upon the previous work by examining these same retirement changes across the *entire* consumption distribution through the application of quantile regression techniques on data from the 1990-2007 Consumer Expenditure Survey. The evidence indicates gradually larger decreases in consumption across the distribution using several measures of consumption, with insignificant changes shown at lower deciles and the greatest drops occurring at the higher deciles.

Keywords: retirement, life-cycle model, household consumption.
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1 Introduction

The basic forward-looking life-cycle model states that consumption should remain smooth through the retirement transition, as the change in income at retirement should be predictable (Modigliani and Brumberg, 1954; Hamermesh, 1984). A body of empirical research has instead found a discontinuous one-time drop in consumption at retirement, contradicting the predictions of the model and thus branding this drop – the retirement consumption puzzle. Subsequent research has instead suggested that there may be no such puzzle (Fisher et al., 2008; Hurst, 2008). Regardless, this collective empirical work has focused on changes in consumption upon retirement at the mean or median, implicitly assuming that these changes, or lack thereof, were uniform across the consumption distribution.

The change in consumption at retirement, however, may differ between higher and lower consumption households. Just as changes in mean earnings are a poor indicator of changes to the distribution of earnings in the United States over the last forty years, the mean changes in consumption at retirement are potentially a poor indicator of what is happening in the tails of the consumption distribution. For example, high consumption households may have been consuming beyond their means while working but find credit less accessible in retirement or income lower than expected. Alternatively, lower consumption households may also be lower income households, and they might also have a drop in consumption at retirement when social security is their lone income source.

This study expands upon the previous work of the retirement consumption puzzle by evaluating whether the change in consumption at retirement is uniform across the

entire consumption distribution. This is achieved through the application of quantile regression techniques on data from the 1990 to 2007 Consumer Expenditure (CE) Surveys. Additional improvements are made to much of the previous work by using five different consumption measures to test the sensitivity of the results, by using actual total consumption instead of food expenditures or imputed total consumption, and by using more recent data.

The findings of this study introduce several new and important facts to the literature. Most importantly, the consumption drop at retirement is found to differ significantly across the consumption distribution. The magnitudes of the consumption drops increase as the level of pre-retirement consumption increases. Surprisingly, no statistically significant drop in consumption at retirement occurs in approximately the bottom fifth of the consumption distribution. This latter finding is interpreted as being in-line with the predictions of the life-cycle model, as there is no evidence of a discontinuous drop in consumption at retirement. On the other side of the distribution, a large and statistically significant drop between 5 to 9 percent occurs for the top two-fifths at retirement. This suggests that a retirement consumption puzzle may still exist among these individuals. The only exception to the overall pattern is for work-related expenditures, which are shown to decrease relatively uniformly across the distribution.

The overall structure of this paper is as follows. Section 2 provides a brief review of the relevant empirical literature. Section 3 describes the CE Survey data, the sample, and all of the variable definitions used. Section 4 presents all of the evidence on the changes in consumption upon retirement. Section 5 concludes the paper with

the discussion and interpretation of the findings.

2 Brief Review

Hamermesh (1984) was the first study to document the discontinuous drop in consumption at retirement. Using an analysis of consumption-to-income ratios, he argued that this discontinuity in consumption may be explained by inadequate savings. Bernheim, Skinner, and Weinberg (2001) estimated the drop in consumption across wealth and income replacement rate quartiles and found that the drop is larger for those with lower wealth and those with lower income replacement rates. They too argued that their findings are consistent with insufficient savings. While these and other studies have used U.S. data, the puzzle has also been documented for several other countries. Banks, Blundell, and Tanner (1998) found a drop in consumption among retiring cohorts in the United Kingdom, only some of which may be explained away. Schwerdt (2005) found a drop in consumption for German households, as well as its negative correlation to income replacement like that shown by Bernheim, Skinner, and Weinberg (2001). Wakabayashi (2008) showed that the puzzle also exists in Japan, which is mostly explained by changes in household size, similar to the arguments made by Banks, Blundell, and Tanner (1998). Lastly, Battistin et al. (2009) utilized a regression discontinuity approach on Italian data and found a drop in consumption at retirement there.[1]

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[1] Please see Attanasio and Weber (2010) for a more comprehensive review of the related literature.

More recent research using U.S. data identified that the puzzle has been, for the lack of a better word – retired (Hurst, 2008). Fisher et al. (2008) showed that the puzzle disappears when using a broader measure of consumption flows instead of just food expenditures, as much of the previous literature has used. Aguiar and Hurst (2009) further bolstered the finding that food is a poor proxy for the consumption of retirees by showing that retirees substitute toward the home production of food and that, despite the drop in food expenditures, the quality and quantity of food consumed stays constant in retirement. Others found that the consumption drop is only for those that retire unexpectedly because of illness, disability, or involuntary unemployment (Haider and Stephens, 2007; Smith, 2006), which also argues against the existence of a puzzle. Blau (2008) developed a life-cycle model that incorporates the uncertainty in the timing of retirement and treats retirement as a discrete event. He found that this modified life-cycle model predicts discontinuous drops in consumption for households that retire unexpectedly.

One common aspect of all of these studies is the focus on the mean or median change in consumption at retirement. The analysis that follows in this study moves beyond these measures of central tendency and tests whether the retirement consumption puzzle is uniform across the consumption distribution. This provides yet another perspective on the existence of the puzzle, which has not been previously addressed.

3 Data and Sample

The Consumer Expenditure (CE) Interview Survey is the premier data source for information on the spending habits of American consumers. It is administered quarterly to a representative sample of households, with the same consumer unit followed for four quarters. For this study, the sample is restricted to those ages 50 to 79 that appeared in all four quarterly interviews and had positive spending on food, nondurable-transportation, clothing and personal care, utilities, and entertainment, along with positive housing services, following Aguiar and Hurst (2009). In order to measure 12-month consumption, these four quarterly consumption values are then aggregated.

While the CE Survey collects data from consumer units, here the unit of analysis is the individual. This conversion is done by disaggregating consumer units using an equivalence scale equal to the square root of family size. This equivalence scale assumes that a two-person household needs to spend approximately 41 percent more than a one-person household for the two consumer units to experience an equivalent level of consumption. This scaled consumption is then multiplied by the consumer unit size and its population weight to serve as an individual's overall weight. Fisher et al. (2008), among others, have also used this type of equivalence transformation. At the end of the next section, the sensitivity of the overall results to the use of an equivalence scale is tested.

Given that the primary focus is on individuals before and after the retirement transition, the sample consists of those aged 50 to 79 years old. These individuals are aggregated in three-year age bands over three consecutive years of data to form

cohorts. For example, individuals aged 50-52 in each year of 1990, 1991, and 1992 are grouped together and are followed through the data as they age together until they reach the ages of 65-67 in 2005-07. The next cohort then begins at ages 53-55 years old in 1990-1992 and is followed until ages 68-70 in 2005-2007. The oldest cohort was 62-64 in 1990-1992 and then 77-79 in the last years of data. Table 1 displays the construction of the cohorts along with sample sizes.

Table 1: Years of Data and Sample Size by Age and Cohort

	Years of Data					Sample Size
Age	Cohort A	Cohort B	Cohort C	Cohort D	Cohort E	Total
50-52	1990-92	.	.	.	.	960
53-55	1993-95	1990-92	.	.	.	1,698
56-58	1996-98	1993-95	1990-92	.	.	2,242
59-61	1999-01	1996-98	1993-95	1990-92	.	3,120
62-64	2002-04	1999-01	1996-98	1993-95	1990-92	4,052
65-67	2005-07	2002-04	1999-01	1996-98	1993-95	3,845
68-70	.	2005-07	2002-04	1999-01	1996-98	3,045
71-73	.	.	2005-07	2002-04	1999-01	2,255
74-76	.	.	.	2005-07	2002-04	1,304
77-79	.	.	.	.	2005-07	510
Sample size	5,347	4,903	4,476	4,237	4,068	23,031

Notes: Authors' calculations of Consumer Expenditure Survey data from 1990 to 2007.

These five cohorts are used because consumption for each is available before and after the age of 65. These cohorts are not the same individuals over time, as there is only one observation per individual in the data. Instead, ages and years

are aggregated to create pseudo-cohorts that are then followed over time. This use of pseudo-cohorts is the only way to obtain sizable cohorts over multiple time periods. Sample sizes are extremely important because this study is interested in estimating effects across the entire consumption distribution, not just measures of central tendency. It may not be possible to use longitudinal studies such as the Panel Study of Income Dynamics (PSID) or the Health and Retirement Study (HRS) to estimate the the retirement consumption puzzle across the entire distribution because of their smaller sample sizes.

A restrictive definition of retirement is used here, counting an individual as retired if she reports no earnings and states retirement as the reason for having no earnings, following Fisher et al. (2008) and other research using the CE Survey. Table 2 shows the percentage retired by age group. At ages 62-64, 31.7 percent of individuals are retired. Just over half of individuals are retired by ages 65-67, and 64.3 percent of individuals are retired at ages 68-70. This pattern is consistent within each of the cohorts. This is also consistent with other data, as the average age at retirement is 62.6 in the Health and Retirement Study (Laitner and Silverman, 2005).

It is a concern that the pseudo-cohorts are changing over time and are therefore not the same types of individuals. To address this concern, Table 2 also displays the median age by consumption decile and retirement status. Individuals with lower consumption are somewhat older than those with higher consumption while working and in retirement. At least along the dimension of age, these individuals do not appear dramatically different before and after retirement.

The last issue is the measure of consumption. The literature has become increas-

Table 2: Percentage Retired by Age and Median Age by Consumption Decile

	Percentage Retired		Non-Retired Median Age	Retired Median Age	Age Difference
50-52	1.3	Avg	61.1	68.4	7.3
53-55	3.0	10th	61.3	69.3	8.0
56-58	5.0	20th	61.1	68.8	7.7
59-61	13.2	30th	61.5	68.6	7.1
62-64	31.7	40th	61.5	68.3	6.8
65-67	51.6	50th	61.3	68.6	7.3
68-70	64.3	60th	60.9	68.1	7.2
71-73	72.1	70th	60.8	68.0	7.2
74-76	76.2	80th	60.7	68.3	7.6
77-79	83.4	90th	60.8	67.8	7.0

Notes: Authors' calculations of Consumer Expenditure Survey data from 1990 to 2007. The median retirement age is calculated as the mean age of the ten percentage point band around the relevant percentile. For example, the 10th percentile is calculated as the mean age among the individuals from the 6th to the 15th percentiles.

ingly aware that the consumption measure matters greatly (Hurst, 2008). Five measures are used to define consumption in this study in order to show the sensitivity of the results to this critical issue. The measures are total expenditures, work-related expenditures, non-work-related expenditures, non-work-related consumption flows, and total consumption flows. All of these consumption measures are expressed in real 2008 dollars.

Total expenditures include all outlays for goods and services. Total consumption flows include all outlays for goods and services like total expenditures, subtracting

out the cost of home-ownership (mortgage principal and interest) and adding back the rental equivalence of the owned home. For renters, total expenditures equal total consumption. Among the five measures, total consumption comes closest to matching the actual consumption of individuals as it captures the flow of services from the owned home. Over 65 percent of older Americans own their home with no mortgage (Fisher et al., 2007), meaning that any expenditure measure tends to understate consumption of housing among home owners.

The work-related expenditures, non-work-related expenditures, and non-work-related consumption flows are constructed following Aguiar and Hurst (2009). Work-related expenditures equal food away from home plus apparel, personal care items, and non-durable transportation.[2] As one would expect work-related expenditures to fall at retirement, work-related expenditures are isolated to determine whether these are driving the overall results. Total non-work-related expenditures include food at home, alcohol, tobacco, utilities, domestic services, and entertainment.[3] The flow of housing services, using rental equivalence, is added to non-work related expenditures to obtain non-work-related consumption flows.

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[2] Food consumption was the lone definition of consumption in much of the early research examining the retirement consumption puzzle. In this study, food consumption is itself divided into food away from home, as a component of work-related expenditures, and food at home, as a part of non-work-related expenditures and non-work-related consumption. [3] Total expenditures includes items that are not in work-related expenditures or non-work-related expenditures, such as medical care expenditures and housing outlays.

4 Changes in Consumption upon Retirement

4.1 At the Mean

The analysis begins by replicating what has previously been done by focusing on the mean effect of retirement upon the various consumption measures. This ordinary least squares (OLS) regression estimation follows existing methodology (Smith, 2006; Fisher et al., 2008; Aguiar and Hurst, 2009) and is of the following form:

$$ln\,(C) = \alpha + \beta \cdot Retired + X'\gamma + \epsilon \tag{1}$$

where $ln\,(C)$ represents the natural log of the equivalent consumption measure as the dependent variable, *Retired* equals the value of one if the individual is retired, and X represents the set of controls. In all of the specifications, vectors of cohort and year binaries are included in X to stay true to the cohort-year framework.[4] Unless otherwise stated, all specifications include demographic binaries spanning gender, race, and marital status as additional control variables.

The coefficient of interest for the OLS estimation, β, can be interpreted as the change in consumption at retirement, represented as a percentage change. It represents the difference in the mean logged outcomes of consumption between the non-retired and the retired. This distinction becomes important when the quantile

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[4] Cohort, year, and age effects cannot be separately identified as the three are perfectly correlated. All results are identical even if a cubic in age and year dummy variables are included without the cohort dummy variables.

methods are introduced in the next section.

Table 3 presents the coefficient on retirement for each of the consumption measures. At the mean, retirement induces a statistically significant decrease in consumption across all consumption measures. The drop in total expenditure at retirement is between 3.7 to 6.2 percent using the weighted and unweighted values, respectively. Work-related expenditures decrease by a larger amount of 5.5 to 9.1 percent while non-work-related expenditures and consumption flows both drop by only 1.5 to 3.0 percent. Total consumption flows also yield a lower drop at retirement as compared with total and work-related expenditures.

These results are consistent with the overall literature in two ways. First, as you broaden the consumption measure to include flows, the retirement consumption puzzle diminishes (Fisher et al., 2008). Second, much of the drop in consumption at retirement is due to changes in work-related expenditures (Aguiar and Hurst, 2009). These themes will be repeated throughout the paper.

The other pattern in Table 3 is that the unweighted results are close to twice as large as the weighted results. In general, unweighted and weighted regressions are both unbiased when regressions are homogeneous across strata. In the presence of heterogeneity, the two expectations will differ (Deaton, 1997). The unweighted estimates will reflect the specific characteristics of the sample, while the weighted set of estimates will be more representative of the population at large. The differing unweighted and weighted results indicate the presence of heterogeneity, which further argues for a quantile regression approach. Both weighted and unweighted results continue to presented throughout the analysis.

Table 3: OLS Regression - Unweighted and Weighted

	Coefficient on Retired (std. error)	
	OLS Unweighted	OLS Weighted
Total Expenditures (TE)	-0.062	-0.037
	(0.008)	(0.009)
Work-Related Expenditures (WRE)	-0.091	-0.055
	(0.011)	(0.011)
Non-Work-Related Expenditures (NWE)	-0.030	-0.016
	(0.007)	(0.008)
Non-Work-Related Consumption Flows (NWC)	-0.029	-0.015
	(0.006)	(0.007)
Total Consumption Flows (TC)	-0.043	-0.022
	(0.008)	(0.009)

Notes: Authors' calculations of CE Survey data from 1990 to 2007. All regressions control for cohort and year binaries and demographic variables. Total number of observations equals 23,031.

4.2 Across the Distribution

The main contribution of this study is to produce the retirement coefficients across the deciles of the consumption distribution. This is achieved by using the quantile regression techniques explained in more detail by Koenker and Bassett (1978), Buchinsky (1998), and Koenker and Hallock (2001). Rather than minimizing the sum of squared deviations around the mean as in OLS, quantile regressions minimize the sum of absolute residuals around the percentile of interest.

Another way to consider the difference between the OLS results and the quantile regression results centers around the interpretation of the coefficient on *Retired*. For the quantile regression, the coefficient estimate on the retired variable represents the difference in the outcomes of equivalized log consumption between the non-retired and retired at the qth quantile of their distributions (i.e. cumulative density functions), as opposed to OLS which provides the difference in the conditional mean of the log equivalized consumption measure at retirement. The quantile regression recognizes the idea that the relationship between consumption and retirement may not be uniform across the distribution. Neither coefficient represents an estimate of the causal effect of retirement on consumption.

More specifically, the first technique utilized in this section is the simultaneous quantile regression. The results of the simultaneous quantile regressions are presented at the 10th through 90th deciles, using the five measures of consumption and controlling for cohort and year binaries as well as demographic variables throughout. This simultaneous quantile regression returns the same coefficients as the independent quantile regression for every percentile, but it produces different standard errors. The simultaneous variance-covariance matrix is estimated by bootstrapping using 50 replicates, which allows for inference between percentile estimates. This allows for explicit hypothesis testing for whether two percentile estimates are equal. For the independent quantile regression, these inference tests are not possible.

Table 4 shows that the change in consumption at retirement differs across the consumption distribution, for four of the five measures. The change becomes more negative moving up the consumption distribution, with the largest change in con-

sumption occurring at the upper end of the distribution. It is the high consumption individuals that experience consumption drops at retirement, while the low consumption individuals experience little to no drop at retirement. In addition, some changes at the low end of the distribution are found to be statistically insignificant, while the statistical significance increases along with the magnitude from the 10th through to the 90th percentile. Examining the first row estimates for the measure of total expenditures, the consumption change moves from a statistically insignificant drop of 3.0 percent at the 10th percentile, to a statistically significant drop of 6.6 percent at the median, to the largest drop of 9.2 percent at the 90th decile.

When only the work-related expenditures are examined following Aguiar and Hurst (2009), the coefficients are almost completely uniform across the distribution. Work-related expenditures are the lone departure from the ever increasing trend in magnitude and significance in the consumption drop at retirement. The estimates are a 7.7 percent drop at the 10th decile, a 7.5 percent drop at the median, and a 8.7 percent drop at the 90th decile. The fact that these results for the work-related expenses of food away from home, clothing and apparel, and transportation are basically identical across the distribution is somewhat expected. When individuals retire, work-related expenses decrease by approximately the same percentage, regardless of level of work-related expenditures, but it did not need to be the case that the percentage changes equaled. Note, however, that these decreases will of course be different in terms of levels, which is an issue revisited later.

In the third row of Table 4, non-work-related expenditures are evaluated. Here, the pattern across the distribution is similar to total expenditures, but the mag-

Table 4: Simultaneous Quantile Regression - Unweighted

	Coefficient on Retired (std. error)								
	P10	P20	P30	P40	P50	P60	P70	P80	P90
TE	-0.030	-0.034	-0.042	-0.057	-0.066	-0.072	-0.072	-0.071	-0.092
	(0.016)	(0.013)	(0.010)	(0.011)	(0.009)	(0.011)	(0.012)	(0.016)	(0.018)
WRE	-0.077	-0.067	-0.085	-0.085	-0.075	-0.078	-0.087	-0.095	-0.087
	(0.025)	(0.016)	(0.016)	(0.014)	(0.012)	(0.009)	(0.010)	(0.013)	(0.021)
NWE	-0.016	-0.008	-0.009	-0.023	-0.022	-0.028	-0.032	-0.034	-0.069
	(0.012)	(0.010)	(0.008)	(0.008)	(0.007)	(0.008)	(0.009)	(0.011)	(0.016)
NWC	0.008	-0.002	-0.006	-0.019	-0.026	-0.031	-0.038	-0.053	-0.068
	(0.010)	(0.006)	(0.006)	(0.006)	(0.006)	(0.008)	(0.007)	(0.007)	(0.011)
TC	-0.019	-0.011	-0.023	-0.039	-0.051	-0.053	-0.057	-0.058	-0.070
	(0.013)	(0.011)	(0.009)	(0.008)	(0.009)	(0.010)	(0.010)	(0.014)	(0.014)

Notes: Authors' calculations of CE Survey data from 1990 to 2007. P10 represents the 10th percentile, and P90 is the 90th percentile. All regressions control for cohort and year binaries and demographic variables. Data are unweighted. Total number of observations equals 23,031. TE = total expenditures; WRE = work-related expenditures; NWE = non-work related expenditures; NWC = non-work related consumption; TC = total consumption.

nitudes of the point estimates are lower. At the 10th percentile, the drop is 1.6 percent, at the 50th percentile it is 2.2 percent, and at the 90th percentile it is 6.9 percent. Because total expenditures in the first row include work-related expenditures, it is expected that the drop in non-work-related expenditures at retirement should be lower. For these particular non-work expenditures, all changes at retirement are statistically insignificant below the 40th percentile. At the 40th percentile and above though, the point estimates are statistically significant. Non-work-related consumption flows in the fourth row follow a very similar pattern to non-work-related expenditures.

Total consumption flows are now turned to in the last row of Table 4. Total consumption yields lower drops in consumption than total expenditures across the distribution, but the rising magnitudes still hold when moving from the low-end to the high-end. Consumption flows drop by 1.9 percent at the 10th, 5.1 percent at the median, and 7.0 percent at the 90th percentile. There are also several bouts of insignificance at the lower percentiles as found with the other non-work measures.[5]

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[5] One other pattern from Table 4 is the u-shaped pattern of the standard errors. The standard errors are higher in the tails of the distribution and are smaller in the middle of the distribution. This shape of the standard error highlights the importance of a large sample size in estimating quantile regressions. With the smaller sample sizes in the PSID and HRS, it may be impossible to obtain precise estimates through the entire distribution.

4.3 Hypothesis Testing

Through the use of the simultaneous quantile regression, hypothesis testing can be conducted to determine whether the percentile estimates are statistically different from one another within a consumption measure. Table 5 presents these hypothesis testing results in the form of p-values comparing the Table 4 estimates to one another. For the simplicity of presentation, only the comparisons using the tail ends of the distribution, the 10th and 90th percentiles, are shown compared with each of the other percentiles.

Table 5 reinforces the patterns seen in Table 4. For total expenditures, the 10th percentile estimate is shown to be statistically different at the five percent level from the estimates at the median and above. Moving to the other end of the distribution, the 90th percentile estimate for total expenditures is statistically different from all of the percentile estimates below the 40th percentile.

For work-related expenditures, the hypothesis that any pair of these coefficients are different from one another is *not* rejected, further reinforcing the uniformity of the drop across the distribution for this particular measure. For non-work related expenditures, the bottom 10th decile is no different from any of the other percentiles with the exception of the 90th. In fact, the 90th percentile is shown to be statistically different from all other decile estimates. The point estimates for non-work expenditures at the 80th percentile and below are from 1.6 to 3.4 percent, while the 90th percentile is more than twice as high as the next highest coefficient (Table 4). This pattern may be interpreted as very little response in non-work-expenditures below the 80th percentile but a large decrease at the 90th percentile. Therefore,

Table 5: Hypothesis Tests Between Coefficients on Retired by Decile

		P-Values from two-sided tests								
		P10	P20	P30	P40	P50	P60	P70	P80	P90
TE	P10	.	0.782	0.395	0.098	0.027	0.015	0.017	0.052	0.014
	P90	0.014	0.007	0.009	0.078	0.179	0.284	0.291	0.188	.
WRE	P10	.	0.599	0.716	0.739	0.943	0.967	0.695	0.489	0.774
	P90	0.774	0.474	0.955	0.955	0.589	0.660	0.984	0.592	.
NWE	P10	.	0.417	0.536	0.591	0.637	0.362	0.228	0.257	0.005
	P90	0.005	0.000	0.000	0.003	0.004	0.004	0.004	0.002	.
NWC	P10	.	0.139	0.135	0.004	0.000	0.000	0.000	0.000	0.000
	P90	0.000	0.000	0.000	0.000	0.000	0.003	0.013	0.157	.
TC	P10	.	0.447	0.775	0.164	0.030	0.035	0.021	0.051	0.006
	P90	0.006	0.000	0.007	0.027	0.160	0.235	0.274	0.252	.

Notes: Authors' calculations of CE Survey data from 1990 to 2007. See Table 4 for the coefficients on retired and their standard errors. Total number of observations equals 23,031.

there is a very small discontinuous drop in expenditures that are not work related, which argues against the existence of a retirement consumption puzzle below the 80th percentile for this sub-category of expenditures.

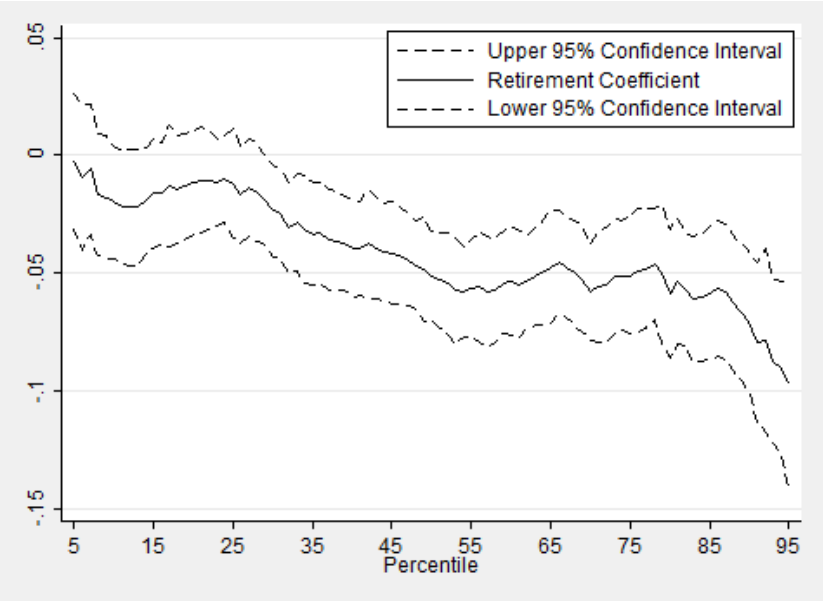
The non-work-related consumption flows estimates are different from one another. The 10th percentile is statistically significantly different from the 40th percentile and above, while the 90th is only similar to the 80th and is different from all other deciles. For total consumption flows, the 10th decile estimate is statistically different from

that of the median and above and the 90th is different from all deciles below the median.

4.4 Graphical Depictions

The following figures fully depict the pattern of changes in consumption at retirement across the distribution. To construct these figures, ninety-one estimates were generated, from the 5th percentile to the 95th percentile.

Figure 1: Retirement Coefficients with 95% Confidence for Total Consumption Across the Distribution



Notes: Authors' calculations of CE Survey data from 1990 to 2007. Figured based on results from Table 4.

Figure 1 shows the pattern for retirement coefficients over the total consumption

distribution with a 95 percent confidence interval. When comparing the retirement effects across the distribution, there is a distinct overall pattern of small and statistically insignificant drops in consumption in the first quarter of the distribution. Moving to higher percentiles, the magnitude of the drops increase from the 25th percentile to the highest drop at the 95th percentile, with a flat portion around the 55th to 75th percentiles.

Figure 2: Retirement Coefficients for Three Consumption Measures Across the Distribution



Notes: Authors' calculations of CE Survey data from 1990 to 2007. Figured based on results from Table 4.

Figure 2 compares the retirement coefficient patterns for total expenditures, work-

related expenditures, and non-work-related expenditures.[6] Total expenditures are shown to follow the pattern of gradually larger decreases in consumption across the distribution. This pattern is similarly followed by non-work-related expenditures, but with smaller magnitudes. Work-related expenditures show large drops in consumption upon retirement that are relatively uniform across the distribution.

4.5 Income and the Consumption Rate

The pattern of an increasing drop in consumption at higher percentiles is not necessarily what one would have hypothesized *a priori*. A good first place to start in seeking an explanation as to what might be generating these results is with income. The CE Survey includes before-tax income of the household.[7] The first two columns of Table 6 report the median before-tax income by total consumption decile by retirement status, and the third column displays the log difference between the two. The 10th percentile exhibits the smallest percentage decrease in before-tax income at retirement, while the 90th percentile exhibits the largest percentage decrease. Focusing on the percentage difference masks the large absolute difference. While the 10th percentile experiences a $4,300 decrease in income, the income of the 90th percentile decreases by over $24,600. The decrease in income for the 90th percentile is larger than the level of income at the 40th decile and below for those retired.

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[6] As noted above, total expenditures are not the sum of work-related expenditures and non-work-related expenditures. [7] Ideally, after-tax income would be used here, but income taxes are not well reported in the CE Survey.

Table 6: Income and Consumption Rate Differences Between Non-Retired and Retired over the Total Consumption Distribution

	Median Before-tax Income			Median Consumption / Income	
	Non-retired	Retired	% Decrease	Non-retired	Retired
P10	18,864	14,528	26.11	0.720	0.922
P20	22,928	16,708	31.65	0.695	0.947
P30	28,360	19,309	38.44	0.670	0.936
P40	31,424	23,020	31.12	0.691	0.970
P50	37,189	25,523	37.64	0.631	0.953
P60	41,054	26,693	43.05	0.682	1.051
P70	44,592	31,917	33.44	0.693	1.001
P80	51,098	33,767	41.43	0.744	1.141
P90	65,851	41,186	46.93	0.752	1.229

Notes: Authors' calculations of CE Survey data from 1990 to 2007. Total number of observations equals 23,031. Income is before-tax. Median before-tax income is calculated as the mean income of the ten percentage point band around the relevant percentile. For example, the 10th percentile is calculated as the mean income among the individuals from the 6th to the 15th percentiles. The percent decrease is estimated as the natural log difference between the two income measures. Equivalent income is used in this table as well as equivalent total consumption.

The last two columns of Table 6 present the median consumption to income ratio at the deciles. Among the non-retired, there is not much difference at the tails in the consumption rate, ranging between 63 percent and 75 percent. The story changes among the retired. The 10th percentile is consuming about 92 percent of before-tax

income in retirement, while the 90th percentile is consuming almost 123 percent of income. Again, this is before-tax income, thereby understating the consumption rate out of after-tax income, presuming the higher income households pay higher taxes.

Table 6 could be consistent with an inadequate savings story or an excessive consumption story. Those at the top of the consumption distribution appear to be spending more before retirement than they can afford to spend in retirement, as evidenced by the last column in Table 6. The highest consumption deciles experience the largest decrease in income and the largest increase in the consumption rate so that they are spending more than their income in retirement.

4.6 Log vs. Level Changes

The previous subsections have analyzed the effect retirement has on consumption. These results were all shown as log changes, showing that the drops are much larger at the top of the distribution in *percentage* terms. In order to see how large the *level* differences in consumption drops are at retirement between the top and bottom of the distribution, these are now calculated and presented in Table 7. This is done using the unweighted simultaneous quantile regression coefficients from Table 4.

The drop in total expenditures was 6.6 percent upon retirement at the median which translates to a drop of $2,351. At the 90th percentile, however, the percentage drop was greater at 9.2 percent but is more than three times as high in terms of levels with a drop of $7,159. The same is shown for work-related expenditures, with the level drop at the median being a third of what it is at the 90th decile, even though the percentage change is uniform. There are even greater differences when looking

Table 7: Log vs. Level Changes

		Value in Dollars		
		P10	P50	P90
TE	Pre-retirement Value	$16,169	$35,622	$77,818
	Retirement Coefficient	-0.030	-0.066	-0.092
	Estimated Change	-$485	-$2,351	-$7,159
WRE	Pre-retirement Value	$2,956	$7,933	$18,748
	Retirement Coefficient	-0.077	-0.075	-0.087
	Estimated Change	-$227	-$594	-$1,631
NWE	Pre-retirement Value	$5,178	$10,518	$21,361
	Retirement Coefficient	-0.016	-0.022	-0.069
	Estimated Change	-$82	-$231	-$1,473
NWC	Pre-retirement Value	$8,547	$15,387	$28,514
	Retirement Coefficient	0.008	-0.026	-0.068
	Estimated Change	+$68	-$400	-$1,938
TC	Pre-retirement Value	$16,283	$34,228	$73,280
	Retirement Coefficient	-0.019	-0.051	-0.070
	Estimated Change	-$309	-$1,745	-$5,129

Notes: Authors' calculations of Consumer Expenditure Survey data from 1990 to 2007. The results use the unweighted simultaneous coefficients from Table 4. Total number of observations equals 23,031. The dollar values are the mean of the consumption measure using the individuals five percentiles above and five percentiles below the stated percentile. For example, the 10th percentile values represent mean expenditures for all working individuals between the 6th and 15th percentiles.

at non-work-related expenditures and non-work-related consumption flows. Here the level drop at the 90th percentile is 4 to 6 times higher in dollars than at the median, though there were also larger differences in percentages for these measures. The level results for total consumption flows follow a similar pattern but do not display as large as a level difference.

4.7 Alternative Specifications

Several alternative specifications are now considered in order to show the robustness of the results from the previous subsections.

4.7.1 Weighted results

The first specification follows Table 4 but uses independent quantile regression for each decile so that weights can be added. As described above, the weights may provide different results in the presence of heterogeneity. Table 8 demonstrates that the pattern of increasing magnitude across deciles is preserved throughout all measures. Interestingly enough, the uniformity in the consumption drops for work-related expenditures now seems to only exist from the 40th percentile and up. The two things that differ with weighting are the magnitudes of the coefficients on retired and the statistical significance of the coefficient across the distribution, especially below the median. These differing magnitudes indicate that the weights do matter, and it reinforces the finding that there is considerable heterogeneity in the change in consumption at retirement.

Table 8: Independent Quantile Regression - Weighted

	Coefficient on Retired (std. error)								
	P10	P20	P30	P40	P50	P60	P70	P80	P90
TE	0.003	-0.013	-0.030	-0.028	-0.046	-0.056	-0.058	-0.047	-0.051
	(0.017)	(0.014)	(0.011)	(0.013)	(0.012)	(0.013)	(0.012)	(0.016)	(0.018)
WRE	0.007	-0.010	-0.039	-0.055	-0.057	-0.059	-0.063	-0.072	-0.058
	(0.028)	(0.018)	(0.014)	(0.015)	(0.014)	(0.015)	(0.016)	(0.016)	(0.020)
NWE	-0.012	0.002	0.002	-0.014	-0.015	-0.019	-0.022	-0.034	-0.051
	(0.014)	(0.011)	(0.010)	(0.010)	(0.010)	(0.010)	(0.011)	(0.014)	(0.018)
NWC	0.020	0.009	0.007	-0.006	-0.020	-0.023	-0.031	-0.045	-0.053
	(0.011)	(0.011)	(0.008)	(0.010)	(0.009)	(0.008)	(0.011)	(0.011)	(0.016)
TC	0.010	0.000	-0.002	-0.019	-0.030	-0.047	-0.043	-0.028	-0.038
	(0.017)	(0.013)	(0.012)	(0.012)	(0.012)	(0.009)	(0.013)	(0.013)	(0.019)

Notes: Authors' calculations of CE Survey data from 1990 to 2007. All regressions control for cohort and year binaries and demographic variables. Data are weighted. Total number of observations equals 23,031.

4.7.2 Only using cohort and year effects as controls

For the next set of alternative specifications, only the total consumption flows (TC) measure is used for the ease of presentation. Consumption flows better represent the actual consumption of the individual because it takes into account the flow of services from the owned home. For comparative purposes, the first row in Table 9 displays the unweighted simultaneous estimates from Table 4. None of the results in Table 9 are weighted. As in Table 8, including weights does not affect the overall

conclusions.

The second row in Table 9 includes only the retirement binary variable and the vectors of cohort and year binaries, but does not control for the demographic variables. Comparing these estimates to the second row specification, it seems that the exclusion of demographic variables lowers the magnitudes of the drops for most of the distribution. Once again, this study finds a statistically significant drop in consumption at the higher parts of the distribution but no such drop in the lower parts of the distribution. In this way, excluding the demographic characteristics fails to affect the main findings.

4.7.3 Consumption instead of equivalent consumption

Under the third specification in Table 9, the dependent variable is left unequivalized and family size binaries are included on the right-hand side instead. Rather than impose a structure for the economies of scale, this specification allows for a non-parametric effect of family size on consumption. The magnitudes of the drops in consumption become more negative across the distribution. However, the pattern of the increasing absolute magnitude of the drop at higher percentiles continues to hold under this alternate way to control for family size.

4.7.4 Adding income

The differences across the distribution may be coming from differences in income, as shown in Table 6. Row 4 of Table 9 modifies row 1 by including the natural log of equivalent income as a control variable, using the same equivalence scale as was used

Table 9: Robustness Tests Using Alternative Specifications - Unweighted

Coefficient on Retired (std. error)									
TC	P10	P20	P30	P40	P50	P60	P70	P80	P90
(1)	-0.019	-0.011	-0.023	-0.039	-0.051	-0.053	-0.057	-0.058	-0.070
	(0.013)	(0.011)	(0.009)	(0.008)	(0.009)	(0.010)	(0.010)	(0.014)	(0.014)
(2)	-0.019	-0.019	-0.010	-0.034	-0.042	-0.056	-0.042	-0.053	-0.067
	(0.013)	(0.012)	(0.012)	(0.011)	(0.011)	(0.011)	(0.013)	(0.013)	(0.015)
(3)	-0.036	-0.019	-0.031	-0.041	-0.053	-0.060	-0.066	-0.065	-0.076
	(0.011)	(0.011)	(0.010)	(0.011)	(0.011)	(0.010)	(0.011)	(0.013)	(0.016)
(4)	-0.017	-0.003	-0.027	-0.034	-0.046	-0.043	-0.047	-0.042	-0.060
	(0.012)	(0.010)	(0.010)	(0.010)	(0.010)	(0.011)	(0.011)	(0.014)	(0.016)
(5)	-0.016	-0.012	-0.028	-0.040	-0.057	-0.065	-0.046	-0.057	-0.053
	(0.017)	(0.012)	(0.011)	(0.013)	(0.012)	(0.012)	(0.023)	(0.012)	(0.016)

Notes: Authors' calculations of CE Survey data from 1990 to 2007. Total consumption is used in each specification. Weights are not used in any specification in this table. Total number of observations equals 23,031 in rows (1) through (4). The sample size in row (5) is 10,942. Row (1) replicates the results from Table 4 for total consumption. Row (2) removes the demographic characteristics from the specification. Row (3) uses the log level of consumption rather than the log of equivalent consumption, and then family size is controlled for non-parametrically on the right-hand side. Row (4) includes the natural log of equivalent income as an independent variable. Row (5) only uses those ages 62-70.

for consumption. Including income generally makes the coefficients smaller, but the pattern of increasing coefficients remains. In results not presented, home ownership status was added with no change in the basic results. Therefore, the inclusion of income and a proxy for wealth cannot remove the observed patterns.

4.7.5 Endogenous retirement

The retirement decision may not be exogenous to consumption. For example, households that retire unexpectedly because of illness or disability may experience a larger drop in consumption. This may be particularly troublesome for those that retire before the normal retirement age.

One way to address the endogeneity of retirement is to restrict the sample by age. By restricting to those ages 62-70, the focus is on those closer to the normal retirement age in the United States. Those in their 50s and 70s may be affecting the results if all factors related to age-related consumption changes are not perfectly controlled for in the standard specification. It also removes from the sample those that retired outside the normal range of ages. One downside is that the sample size falls to 10,942 individuals. Other than the change in sample, the specification is identical to that of row (1). These new results are presented in row (5) of Table 9. The pattern of coefficients is basically identical between row (1) and row (5), suggesting that the inclusion of fifty-year olds and seventy-year olds does not affect the conclusions.

A cubic in age is also included in place of the cohort effects to better control for the age-consumption profile. The results are quantitatively and qualitatively

similar to those presented in Table 9 and are available upon request. Finally, an instrumental variables (IV) model is estimated along the lines of Chernozhukov and Hansen (2005). Retirement is instrumented for using a cubic in age. The model is estimated, but the results are not presented here. The standard errors are so large that the point estimates may have little to no meaning. For what it is worth, the pattern of increasing coefficients is maintained in the IV estimates and is even more dramatic than any of the previously presented results. Given the large standard errors, however, it is difficult to discern anything further.

5 Discussion

Previous research had focused on changes to consumption at retirement at the mean or median and did not investigate whether the retirement consumption puzzle differed across the consumption distribution. This study remedied that oversight by applying quantile regression techniques to consumption data from the 1990-2007 Consumer Expenditure Survey. The so-called retirement consumption puzzle is found to differ across the consumption distribution using multiple measures of consumption, with the exception of work-related expenditures, which drop uniformly across the distribution. The change in consumption at retirement in the bottom fifth of the distribution is small and statistically insignificant, while the drop in the top two-fifths ranges from 5-9 percent and is statistically significant. The quantile regression results confirm two other findings from this literature: the retirement-consumption puzzle begins to disappear as the consumption measure is broadened and that much

of the drop in consumption upon retirement is due to the changes in work-related expenditures.

An important question still remains: how should all of this new evidence be interpreted within the context of a life-cycle model and within the context of using consumption to measure well-being? At the bottom of the distribution, the evidence suggests that the puzzle does not exist, which would be consistent with a standard life-cycle model. Given that, the findings for the lower quarter of the distribution have no bearing on whether retirement consumption is adequate in a normative sense. All that can be determined is that there appears to be little to no drop in consumption in retirement for individuals in at least the bottom 20-30 percent of the consumption distribution. This suggests that the retirement social safety net, along with any personal savings, protects the consumption of those at the lower end of the consumption distribution. Using consumption as a measure of well-being, the results suggest that retirement does not appear to negatively affect the well-being of those in the bottom of the distribution.

At the middle and high consumption percentiles, however, the discontinuous drop in consumption still exists. A basic life-cycle model would argue that these findings warrant the cognomen – retirement consumption puzzle. Assuming retirement is a predictable shock to income, life-cycle consumers should smooth consumption over this shock. One of the main explanations for the existence of the puzzle in previous research is inadequate savings, and the results of this study would not contradict that conclusion. It is shown that those in the top of the consumption distribution experience the largest decreases in income at retirement and that they are consuming

beyond their before-tax income in retirement.

One problem is that an explanation is needed for the joint finding that there is a discontinuous drop in consumption at retirement for high consumption individuals but not for low consumption individuals. Inadequate savings is unlikely to be unique to those in the top of the consumption distribution. Blau (2008) potentially explains the results found in this study within the context of a modified life-cycle model. His model incorporates uncertainty in the timing of retirement and finds that this modified life-cycle model predicts discontinuous drops in consumption for households that retire unexpectedly because of the large negative income shock due to unexpected retirement. Presumably this negative shock is larger for higher income and therefore higher consumption individuals. A larger income shock should induce a larger consumption response, thus generating the distributional results presented here. But, this interpretation relies on a large portion of individuals having an unexpected retirement.

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